MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND (ADJUSTMENT TYPE, LEAD-SCREW ACTUATED), ESTABLISHED RELIABILITY STYLE RJR24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RJR24, established reliability, adjustment type, lead-screw actuated, nonwire-wound, variable resistors. This style is available in characteristics C, F, and H.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification. The following specification forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of this document shall be that listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-39035 - Resistors, Variable, Nonwire-Wound (Adjustment Type), Established Reliability, General Specification For.

(Copies of the specification required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39035, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

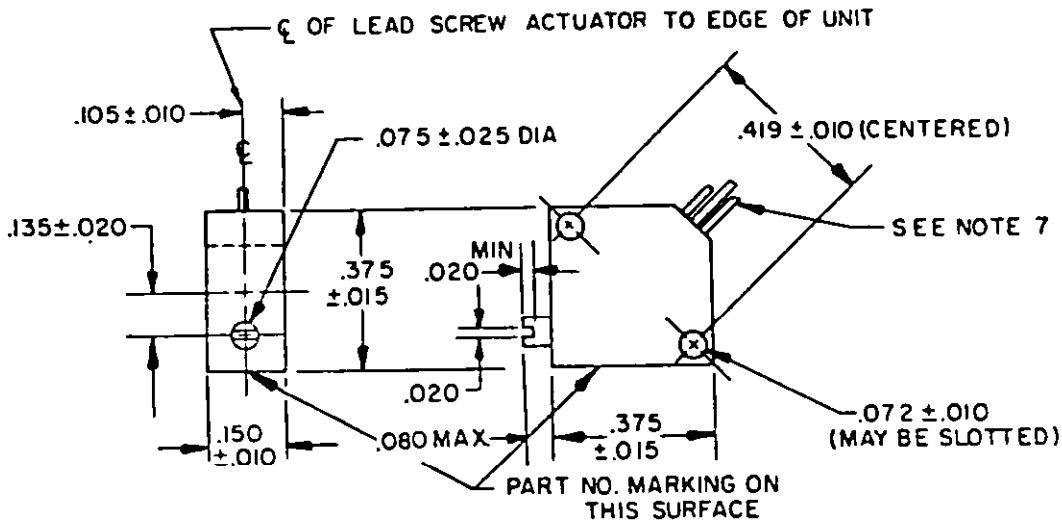
* 3.3 Terminals. Characteristics C, F, and H are available with A-, B-, L-, P-, W-, and X- type terminals.

3.4 Power rating. The power ratings shall be 1/2 watt for all characteristics.

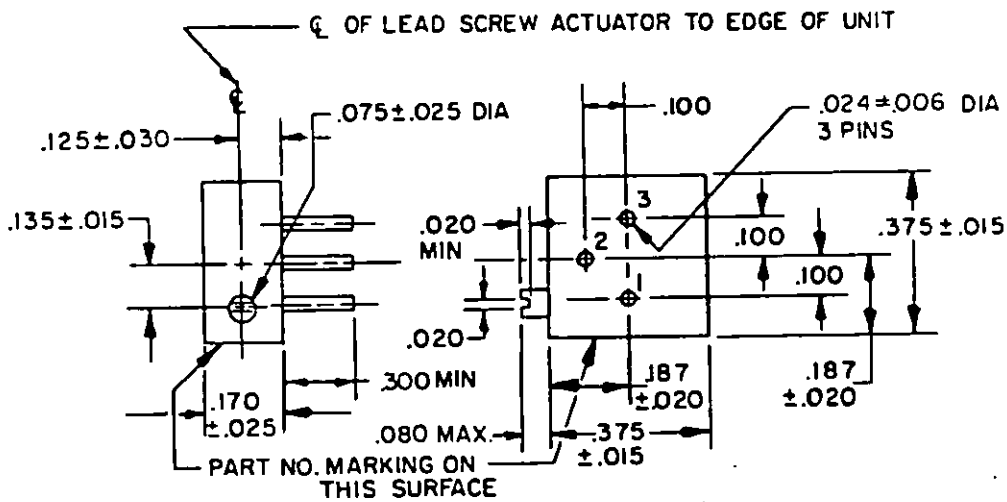
3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be address to: Aeronautical Systems Division, ASD/ENES, Wright-Patterson AFB, OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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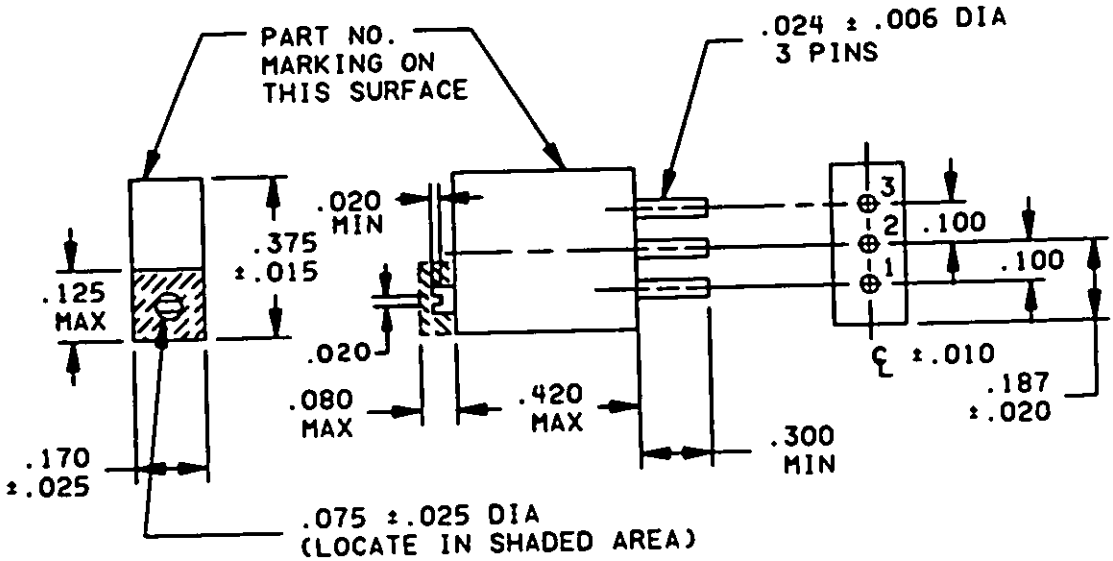
TERMINAL TYPE L



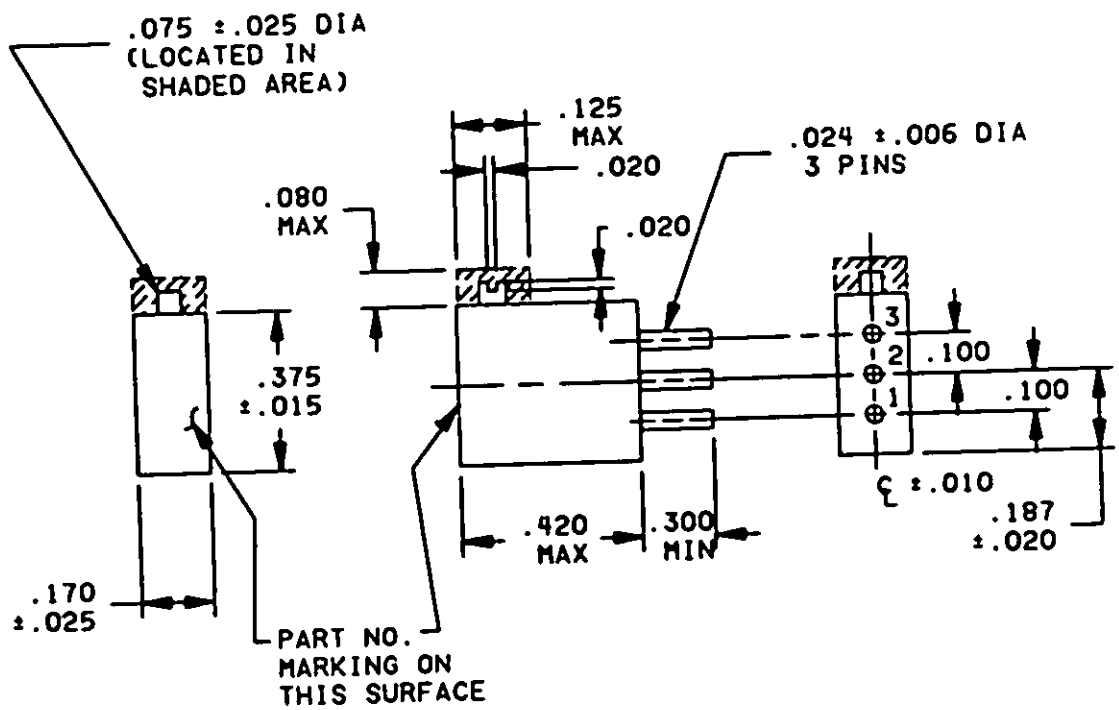
TERMINAL TYPE P

FIGURE 1. Style RJR24 resistors.

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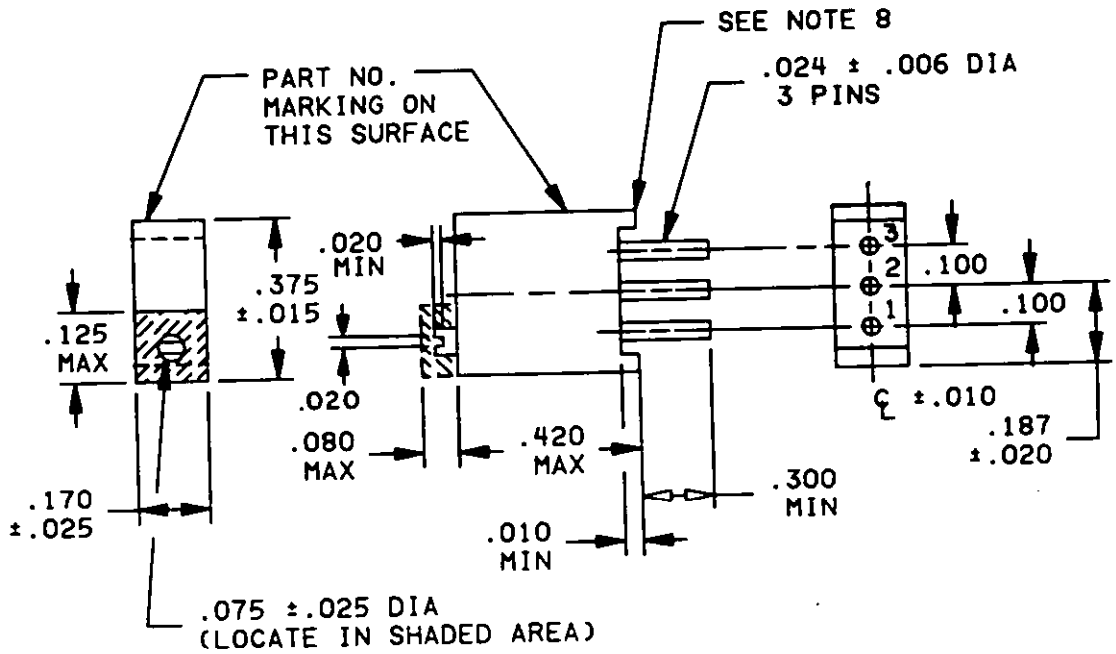
TERMINAL TYPE W



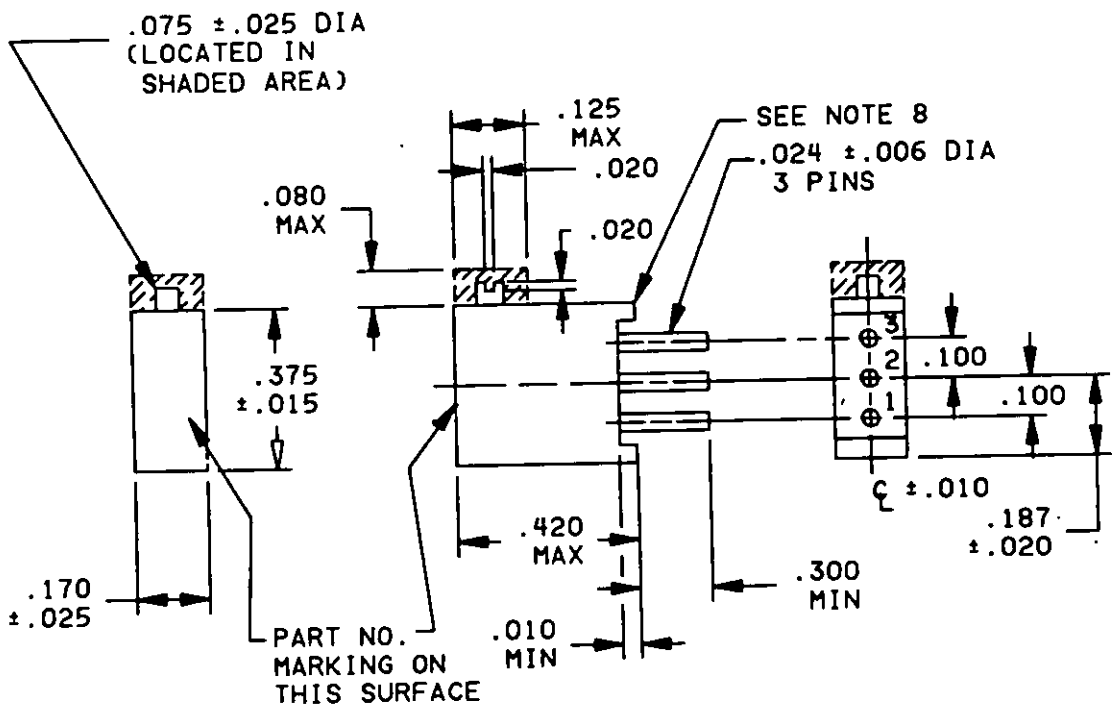
TERMINAL TYPE X

FIGURE 1. Style RJR24 resistors - Continued.

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TERMINAL TYPE A



TERMINAL TYPE B

FIGURE 1. Style RJR24 resistors - Continued.

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Inches	mm
.006	0.15
.010	0.25
.015	0.38
.020	0.51
.024	0.61
.025	0.64
.030	0.76
.072	1.83
.075	1.91
.080	2.03
.100	2.54
.105	2.67
.125	3.18
.135	3.43
.150	3.81
.170	4.32
.187	4.75
.300	7.62
.375	9.53
.419	10.64
.420	10.67

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
5. The entire slot of the actuating screw must be above the surface of the unit.
6. For types P, W, X, A and B, normal mounting means is by use of pins only.
7. The three leads shall be of stranded wire, AWG size 28 to 30, having a minimum length of 6.00 (152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250 ±.062 (6.35 ±1.57 mm) from the end and color coded.
8. Design of standoff is optional.

FIGURE 1. Style RJR24 resistors - Continued.

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3.6 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum, and 30 turns maximum.

3.7 Operating torque. Operating torque shall be a maximum of 5 ounce-inches.

3.8 Maximum voltage. Maximum rated ac or dc working voltage shall be 300 volts.

TABLE I. Nominal resistance value and maximum rated ac or dc working voltage.

Nominal resistance value	Maximum rated ac or dc working voltage per characteristic C, F, and H
Ohms	
10	2.23
20	3.1
50	5.0
100	7.0
200	10.0
500	15.8
1,000	22.3
2,000	31.6
5,000	50.0
10,000	70.7
20,000	100
25,000	111
50,000	158
Megohms	
0.10	223
0.25	300
0.50	300
1.0	300

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39035, and as specified herein.

4.2 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts at atmospheric pressure, and 350 volts at reduced barometric pressure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39035.

6. NOTES

6.1 Notes. The notes specified in MIL-R-39035 are applicable to this specification.

6.2 Weight. The maximum weight is .00286 pound (1.3 grams).

6.3 MIL-R-22097 substitution data. Resistors of this specification regardless of their failure rate designation are substitutes for resistors of the same resistance value, tolerance, terminal characteristic and resistance temperature characteristics specified in the inactivated specifications MIL-R-22097/3 and MIL-R-22097/4.

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-39035/2E	RESISTORS, VARIABLE, NONWIRE-WOUND (ADJUSTMENT TYPE, LEAD-SCREW ACTUATED), ESTABLISHED RELIABILITY STYLE RJR24

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

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